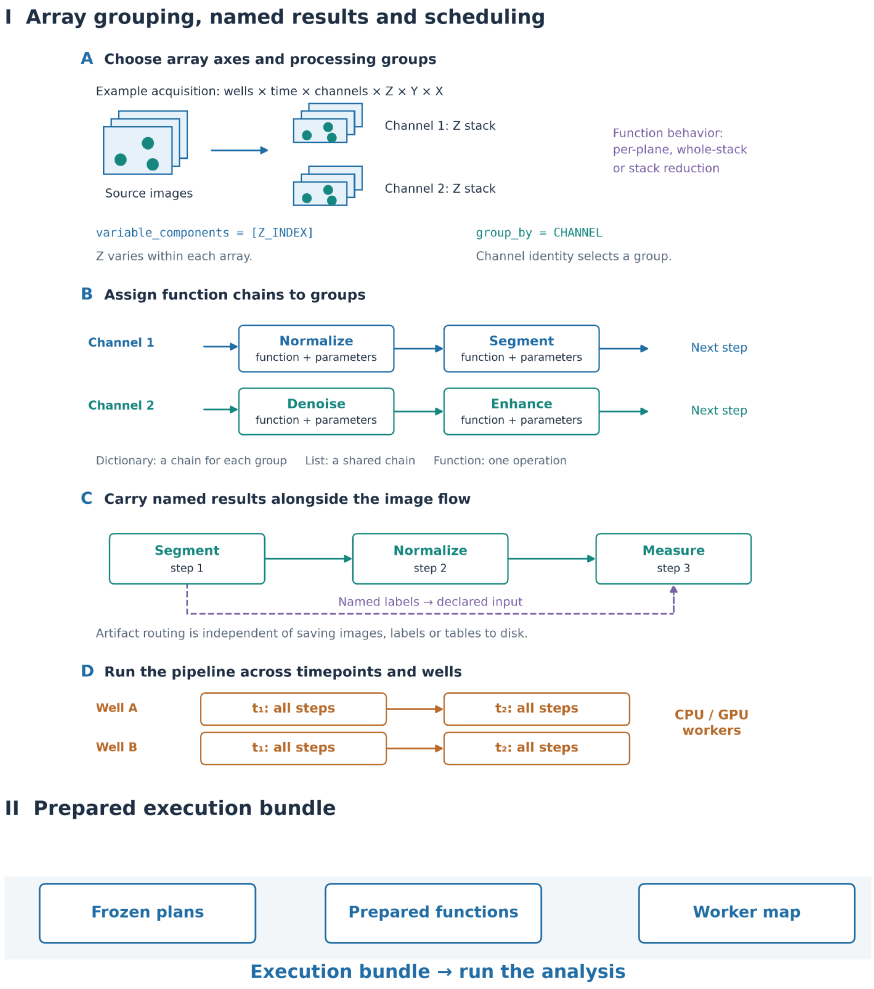
OpenHCS supplementary material

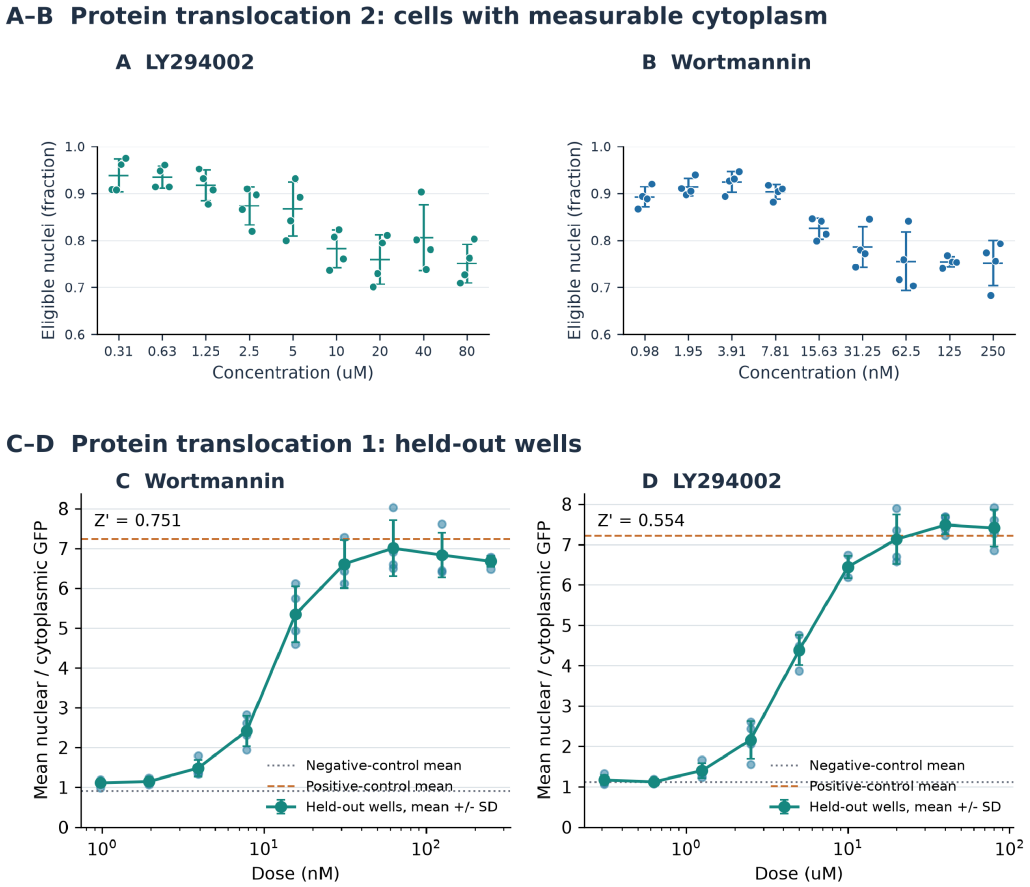
This supplement gives additional results, measurement definitions and reproducible methods. Linked source records provide the original pipelines, instructions and data.

Supplementary Figure 1. Runtime composition and preparation



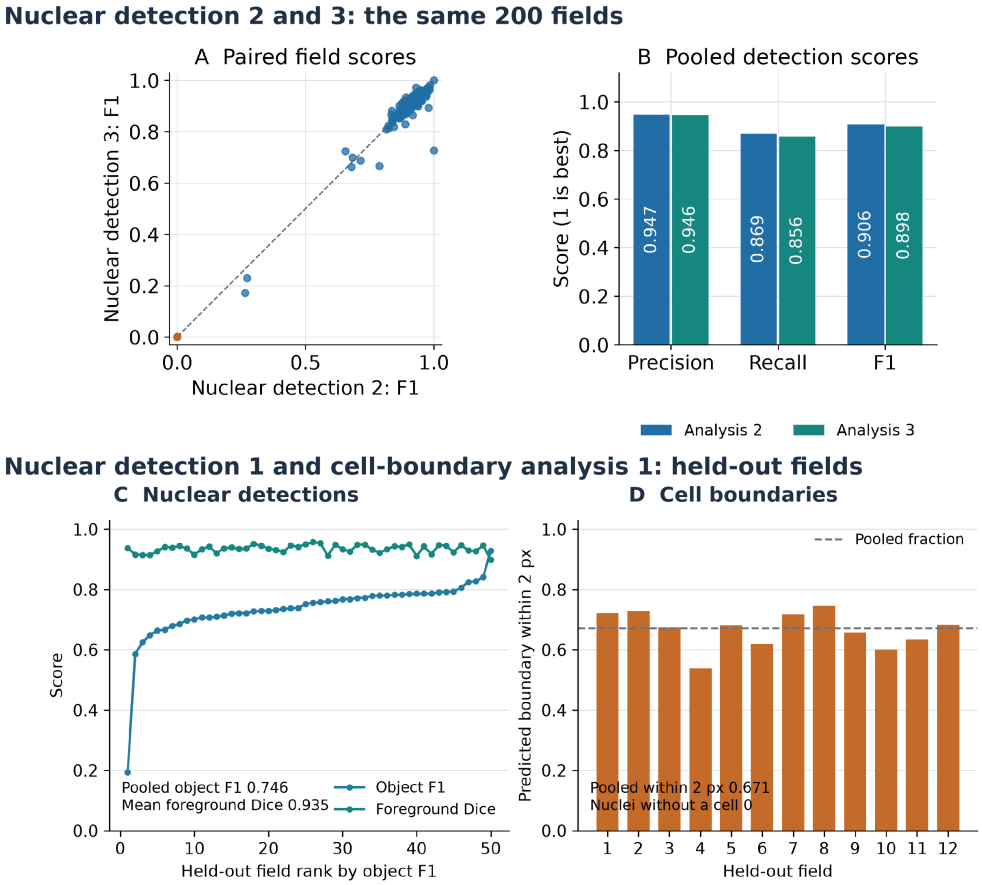
(I, A–D) Array axes and groups supply image views to per-plane, stack or stack-reduction functions. Named labels remain available to later measurements; sequential timepoints complete in order while wells can run in parallel. (II) Compiled plans, prepared functions and a worker map form the execution bundle. The figure assembly and interface records link the source artwork; main Figures 1–2 show the native workflow and process topology.

Supplementary Figure 2. Translocation response and compartment eligibility



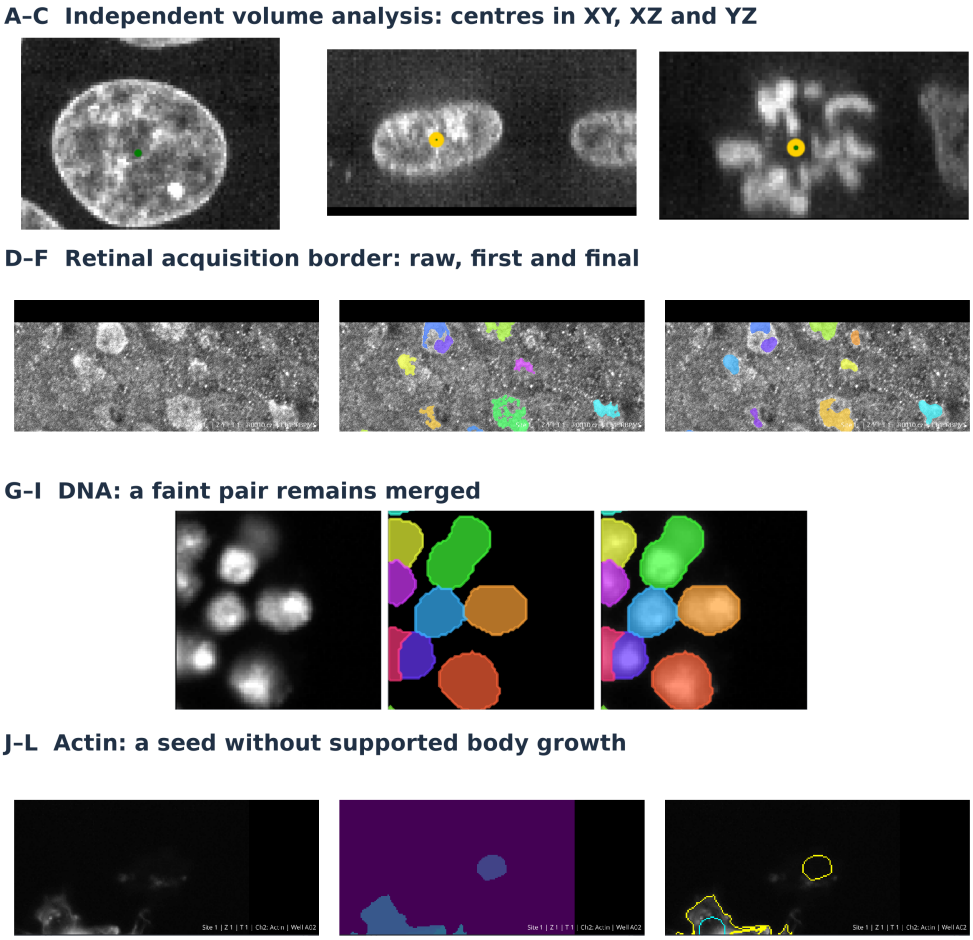
(A–B) Protein translocation analysis 2: fraction of nuclei with measurable cytoplasm in 96 BBBC013 wells, including four development wells. Dots show four wells per dose; marks and whiskers show their mean and sample standard deviation (variation among wells). Main Figure 5E shows the response curves. (C–D) Protein translocation analysis 1: mean cell-level GFP ratios on held-out wells, with four wells per control condition. Z′ measures separation between positive and negative controls relative to their variation; higher is better, and values above 0.5 indicate an excellent assay. Analyses 1 and 2 use different well summaries: mean GFP ratio and median log2 ratio, respectively. Supplementary Data 7–8 give the methods and source data.

Supplementary Figure 3. Nuclear-instance evaluation and object separation



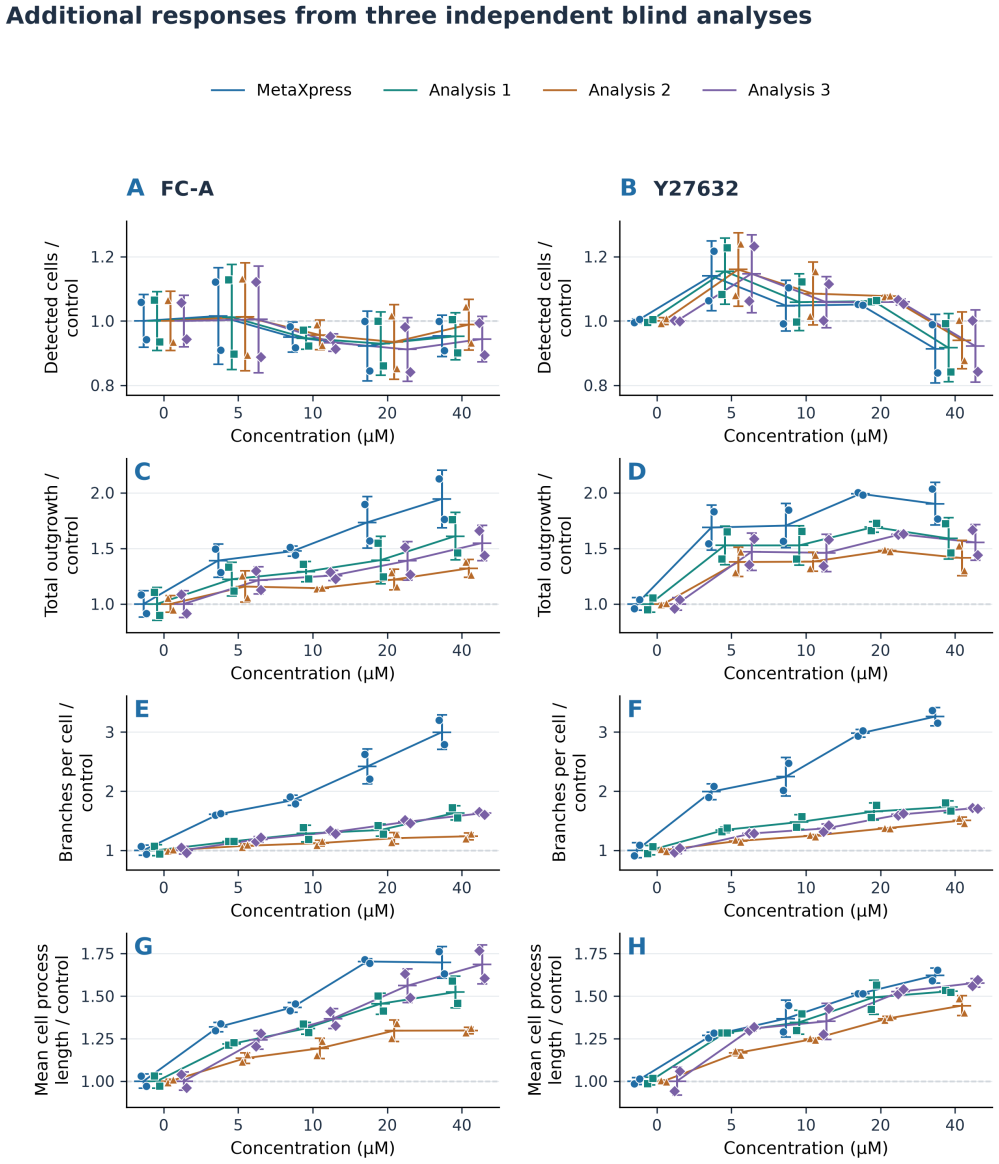
(A–B) Nuclear detection analyses 3 and 2 on 200 BBBC039 fields, with pooled F1 of 0.898 and 0.906, respectively. F1 balances missed and false detections; 1 is best. Matching required intersection over union ≥ 0.5: overlap covering at least half the total area of the two outlines together. (C–D) Nuclear detection analysis 1 on 50 held-out fields and cell-boundary analysis 1 on 12 held-out fields. Each used four development fields. Dice measures area overlap, with 1 indicating identical areas. Boundary support is the fraction of predicted boundary pixels within two pixels of a manual outline; it does not measure whether every manual border was found. Supplementary Data 7–8 provide counts and per-field scores.

Supplementary Figure 4. Independent morphology checks and remaining ambiguities



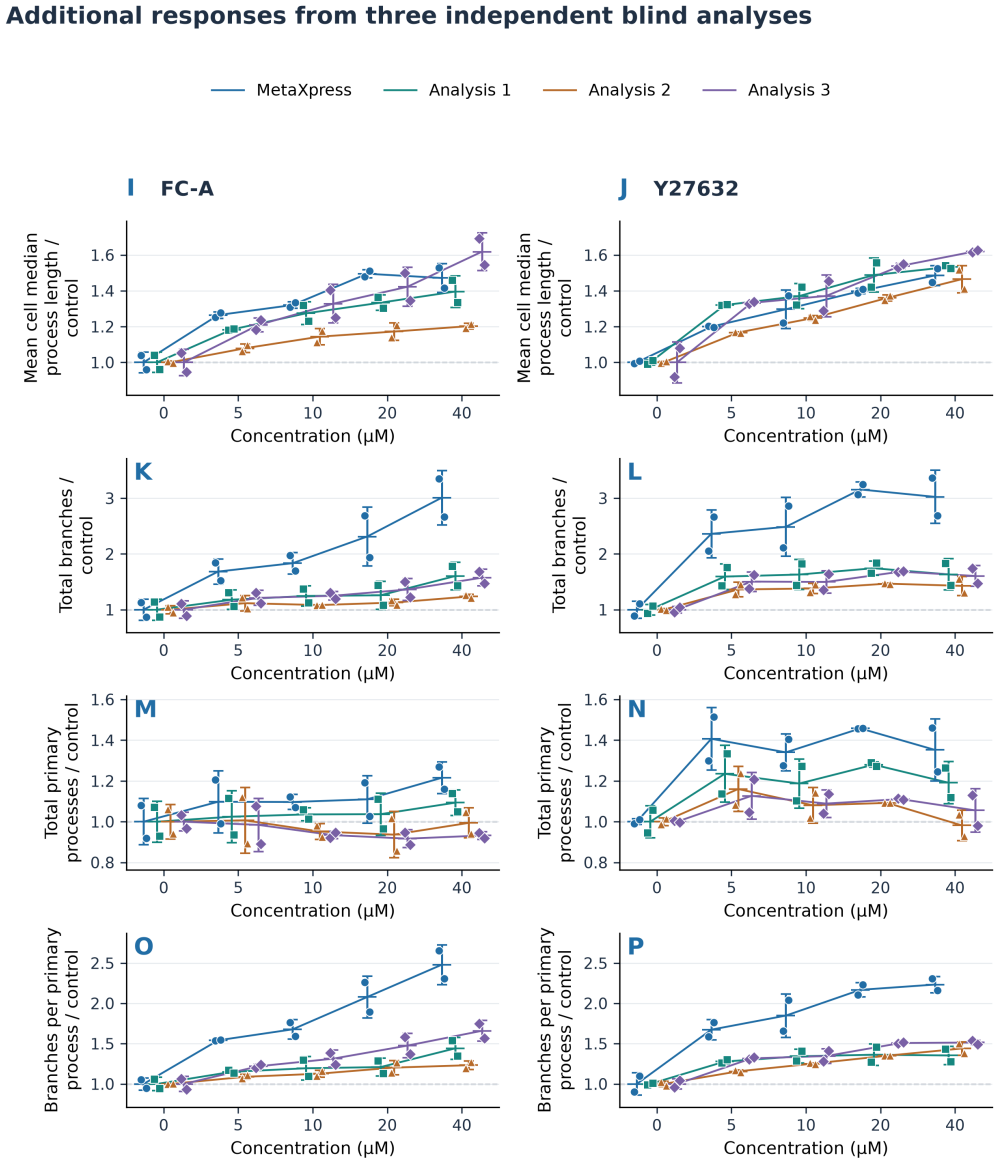
(A–C) Nuclear centres from an independent volume analysis in XY, XZ and YZ. (D–F) Retinal border region shown raw, with first and final outlines; a possible split remains at display window 0–42, gamma 1. (G–I) A faint DNA pair remains merged at window 0–151. (J–L) A nuclear seed lacks actin-supported body growth at window 0–104, gamma 1. The panels show separate trials; colours identify objects within each image. Supplementary Data 8 links the source captures and matching records.

Supplementary Figure 5. Laboratory treatment responses compared with MetaXpress



(A–B) Detected cells; (C–D) total outgrowth; (E–F) branches per cell; (G–H) mean cell process length. Each blind agent completed the same 180 fields in 20 wells. Curves use each drug's own DMSO mean; dots are two technical wells per dose per analysis or method, with sample SD whiskers. Analysis curves are not pooled. Paired drug panels share vertical limits. Main Figure 4A–B shows mean outgrowth per cell.

Supplementary Figure 5, continued. Process length and branching definitions



(I–J) Mean cell median process length; (K–L) total branches; (M–N) total primary processes; (O–P) branches per primary process. Normalization, technical wells and whiskers are as above. Total quantities are mean field totals, not deduplicated whole-well measurements. OpenHCS branch/process values are means of field ratios; MetaXpress values are ratios of exported well totals. These remain distinct algorithm-defined endpoints.

Independent laboratory analyses finalized before scoring

Three fresh gpt-6.1-sol agents (high reasoning effort, OpenAI provider) received the identical task brief, acquisition information, MCP and fixed version of the OpenHCS instructions. The trial record identifies the software and instruction versions, planned development and evaluation images, finalized methods and outputs, and original supporting files. Analyses 1–3 use the same numbering as main Figure 4. Each completed all 20 wells with nine sites per well. Reference and scientist-assisted measurements were opened only after all three agents' outputs were finalized. No agent received score feedback or treatment identities.

The files below preserve the original methods, agents' assessments and unsuccessful attempts without changes after evaluation:

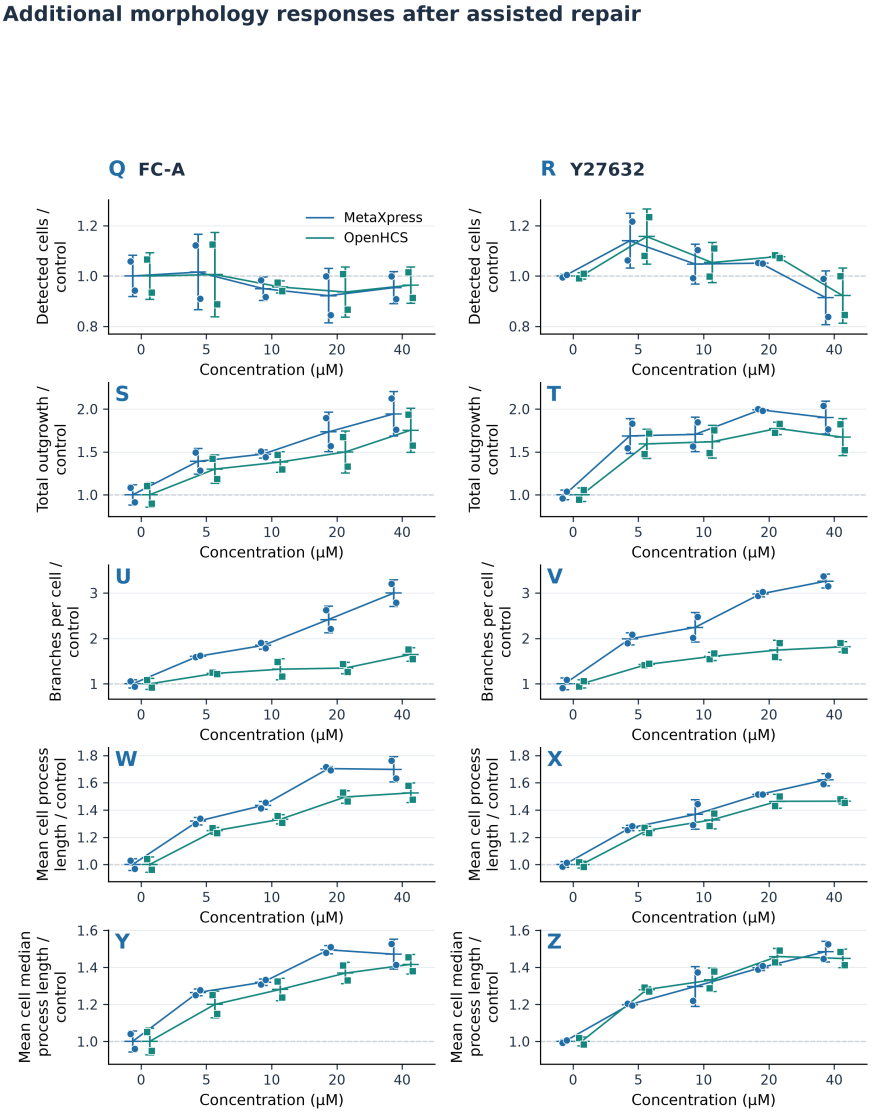
- Analysis 1: pipeline, method freeze, author report, matched wells and effects.
- Analysis 2: pipeline, method freeze, author report, matched wells and effects.
- Analysis 3: pipeline, method freeze, author report, matched wells and effects. Its registered custom sources for channel-selective smoothing, compact native output and coded-well summaries are included.

All three recovered increased mean and total outgrowth at every nonzero dose of both drugs and the same cell-count response directions as MetaXpress. Two were close to the assisted response magnitudes; Analysis 2 gave weaker increases. At 40 μM, mean-outgrowth-per-cell fold changes for analyses 1–3 were 1.646/1.323/1.601 for FC-A and 1.694/1.487/1.660 for Y27632. The assisted values were 1.766/1.765, and MetaXpress's were 1.977/2.053. These are three independent analyses of the same two technical wells per dose, not additional biological replicates.

Reviewing raw images beside the tracings showed recovery of substantial neurite shafts and branches, but did not establish complete tracing or anatomically correct branch counts. After inspecting held-out images, Analysis 3 reported that its measurements should not be interpreted as complete neuron reconstructions or anatomical branch counts. Treatment-effect recovery was evaluated separately, after the analysis was finalized. The linked report also records an operational protocol deviation: execution began before the computer-resource check completed. The report preserves the sequence; treatment identities and reference measurements remained withheld until finalization.

Each agent’s input and output record contains checksums identifying the private treatment key, workbook, pipeline and original CSV files. The records for Analysis 2 and Analysis 3 use the same sampling and averaging rules. Tables pair the same fields and report differences and ratios between assisted and blind results. The assisted analysis is unchanged and used a different software version. Undefined measurements are not replaced with zero. Process-length means in these comparison tables include all detected cells, with zero for cells without processes; the agents’ separate summaries included only cells with processes.

Supplementary Figure 5, continued. Scientist-assisted analysis



(Q–R) Detected cells; (S–T) total outgrowth; (U–V) branches per cell; (W–X) mean cell process length; (Y–Z) mean cell median process length. Twenty matched wells were analysed after assisted repair. Each curve is normalized to its own DMSO mean; dots show two technical wells per dose and whiskers their sample SD. Mean outgrowth per detected cell is included in the linked six-endpoint sheet.

Treatment evaluation and endpoint definitions

For the laboratory neurite comparison, we applied the frozen recipe settings to nine fields in each of 20 FC-A and Y27632 wells using the recorded software

version. An evaluation-only source key linked coded inputs to physical wells. Each well's endpoint was the unweighted mean of nine field-level measurements: outgrowth per detected cell, cell count, total outgrowth, branches per cell, and mean and median process length. For the last two endpoints, we first averaged the corresponding per-cell values within each field, including zero-growth cells. Overlapping fields were not deduplicated; total outgrowth therefore denotes a mean field total, not unique whole-well length. Existing MetaXpress well exports supplied the comparison response, not manual tracing truth. This fixed-recipe transfer was separate from autonomous pipeline authoring.

We retained these predictions when subsequent visual inspection identified shafts included in soma masks, spurious soma-edge branches and supported paths lost during tracing. Repairs informed by those external observations were evaluated separately as assisted development, not autonomous authoring. The comparison retained the same channels, pixel calibration, field sampling and well aggregation. Matched raw-image, soma-mask and path views were used to check recovered shafts and genuine branches alongside unsupported routes; agreement with MetaXpress fold changes was not a tuning criterion.

The commercial export contains 120 well summaries from two plates. The assisted analysis used DAPI w1 and calcein w2, XY spacing 1.3556 µm per pixel, neurite response 30 and candidate admission 0.03. Changes to neurite assignment and branch counting left detected-cell counts unchanged in every field of the 180-field evaluation. The analysis record identifies the software and settings.

Each FC-A and Y27632 concentration has two technical wells. Y27632 is labelled Y27 in the export. The six-endpoint sheet includes mean outgrowth per detected cell and the endpoints in Supplementary Figure 5.

Every endpoint is divided by the same drug curve's zero-dose DMSO mean. Dots show the two technical wells at each concentration. Marks and whiskers show their mean and sample standard deviation after division by the observed control mean; they do not propagate uncertainty in that denominator or represent confidence intervals. Dose positions are equally spaced for display, not a fitted concentration–response model. Paired treatment panels use the same vertical scale. The plot and numerical tables are generated from the same well measurements.

The commercial endpoint is an existing well export whose exact site weighting and software settings are not retained. The MetaXpress 6 Neurite Outgrowth guide defines total outgrowth in micrometres with diagonal-length correction,

primary processes as outgrowths attached to cell bodies, and branches as branching junctions rather than daughter-process counts. It does not specify how the module distinguishes crossings from branches. These documented definitions do not establish identical segmentation or topology for the retained export; absolute lengths and counts are not treated as equivalent. Within-method ratios avoid a constant unit conversion but do not remove those measurement differences. Neither method is manual ground truth, and two technical wells do not establish biological replication or significance.

OpenHCS process lengths describe the connected paths assigned to a soma-adjacent root, including daughter branches, rather than individual segments between graph junctions. Branches require at least three same-neuron paths and distinct admitted image arms around the junction. Paths sharing one image corridor and graph stars confined within a foreground cap do not add branch events; resolved crossings are not automatically branches. These definitions specify the OpenHCS measurements without asserting that the commercial algorithm uses identical topology or aggregation.

The joined physical-well table and treatment-effect table retain well identities, means, sample standard deviations, counts, raw deltas, fold changes and fractional-change differences. Cell-count changes accompany outgrowth to expose denominator changes without inferring toxicity. The source hashes identify the exact submitted pipeline, source key, commercial export and all 180 native summaries. The comparison script processes tables only and does not tune images. Like total outgrowth, averaged field counts do not denote unique whole-well neurons. Unselected wells are not filled with zero. The original frozen comparison is retained separately for before/after evaluation; it was not overwritten.

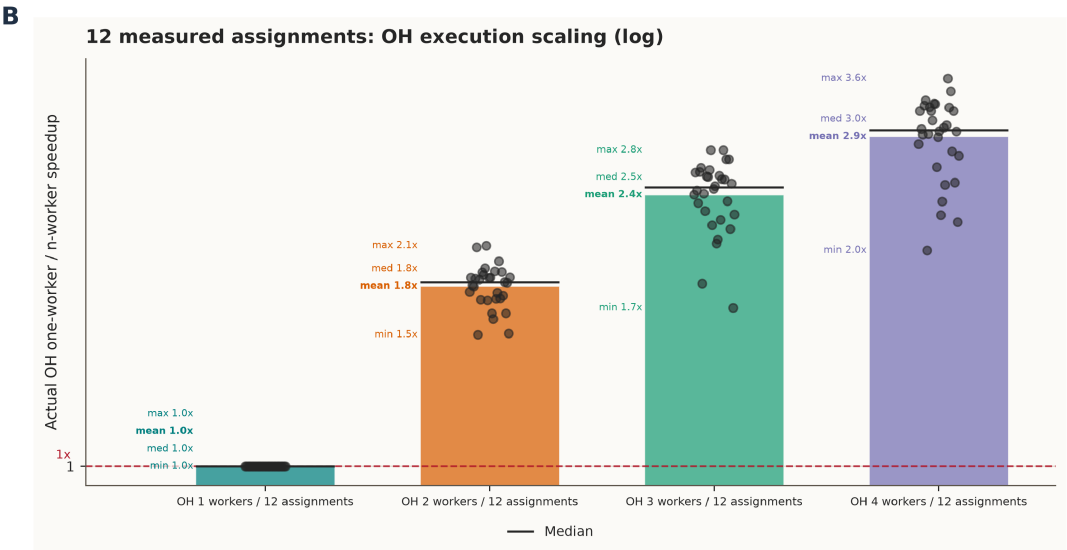
Additional branch and primary-process totals and their treatment responses test whether smaller estimated branching fold changes are explained by the cell denominator. At 40 μM, total-branch fold changes are 1.65 versus 3.01 for FC-A and 1.73 versus 3.03 for Y27632 (OpenHCS versus MetaXpress). Branches per primary process likewise give smaller estimated fold changes: 1.31 versus 2.48 and 1.29 versus 2.23, respectively. The difference therefore persists without a cell-count denominator. OpenHCS totals are means of field totals, and its branch/process endpoint is the mean of the nine field ratios. The commercial comparison uses ratios of exported well totals; its internal site weighting and primary process definition remain unspecified. These diagnostic ratios are not proof of equivalent topology or of which method is more accurate. Their source

record identifies the same completed assisted evaluation, not an additional autonomous trial.

Assisted transfer results

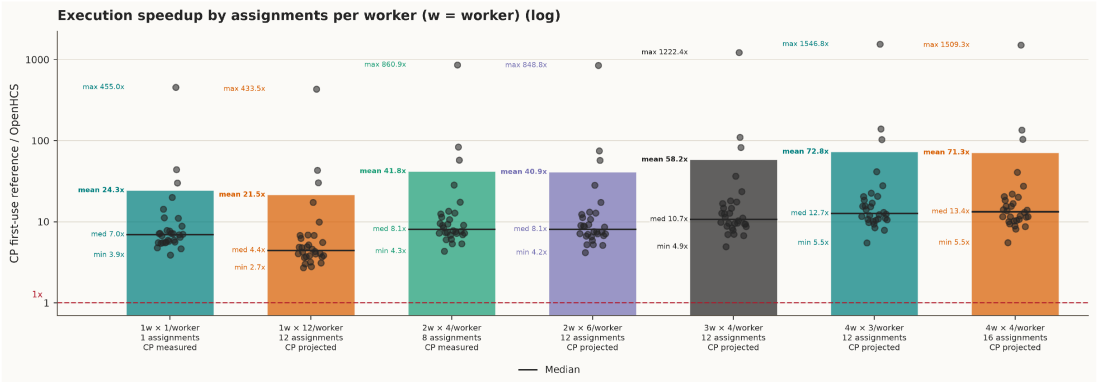
Following externally informed repairs, analysis of all 180 fields in 20 matched wells recovered increasing outgrowth-per-cell responses to FC-A and Y27632 (Supplementary Figure 5Q–Z). At 40 μM, OpenHCS fold changes were 1.77 and 1.76, respectively, versus MetaXpress's 1.98 and 2.05. Total-outgrowth estimated fold changes were also smaller. Branches-per-cell fold changes were 1.65 versus 3.00 for FC-A and 1.82 versus 3.26 for Y27632. OpenHCS control branch counts per cell were approximately 10–11% higher, but treated counts were 38–39% lower. Cell counts were unchanged in all 180 fields after the changes to neurite assignment and branch counting; the difference persisted in total branches and branches per primary process. Control excess alone therefore does not explain the smaller estimated branching fold changes. Remaining branch recall, assignment and measurement-definition differences cannot be separated without spatial ground truth. Both methods recovered concordant positive outgrowth responses with different measured magnitudes. This is an assisted-development evaluation, separate from autonomous authoring.

Supplementary Figure 6. Measured worker scaling and serial-CP comparisons



Measured OpenHCS execution scaling with twelve repeated samples held fixed and 1, 2, 3 or 4 workers. Each point is one workflow's one-worker time divided by its time at the indicated worker count; all thirty workflows appear in each configuration. Times are medians of three repetitions after warmup. Bars show arithmetic means, black lines medians and the dashed line equal runtime. These measurements require no CellProfiler projection.

Supplementary Figure 6, continued. Serial CellProfiler reference

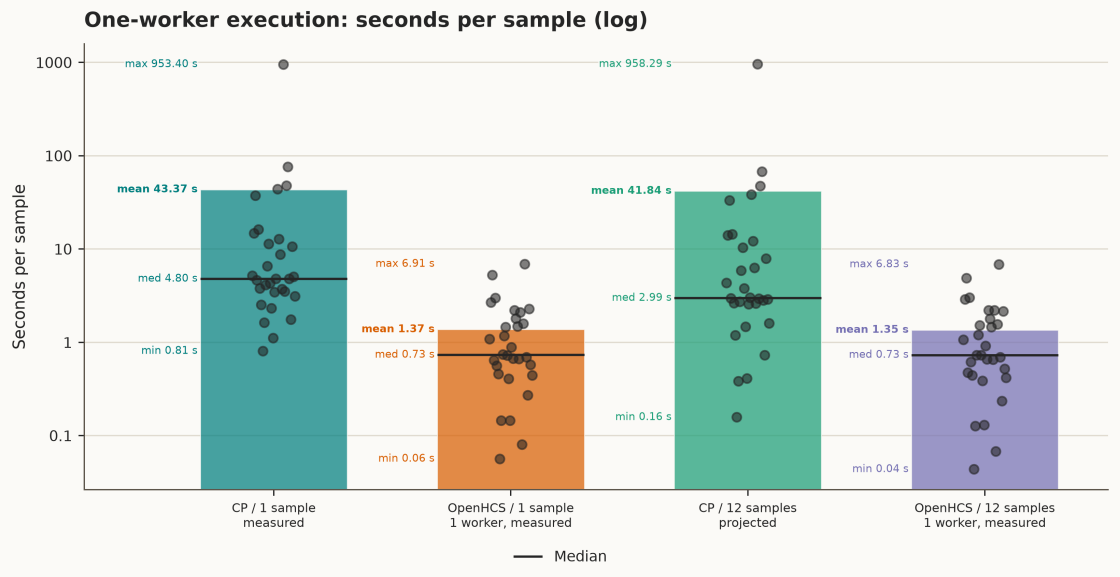


Each configuration contains thirty workflow points; bars show arithmetic means and black lines medians. The labels give OpenHCS workers (w), assignments per worker and total assignments. CellProfiler remains serial: its 1- and 8-assignment first-use-inclusive batches were measured; 12- and 16-assignment references were projected from the measured 8-assignment batch and warmed per-assignment rate. OpenHCS times are medians of three repetitions after warmup. The dashed line marks equal runtime. Repeated assignments are computational replicates, not independent biological samples. Unlike main Figure 6C, these ratios do not match CellProfiler's process count to OpenHCS workers. Main Figure 6B shows the measured single-sample and per-sample batch comparisons.

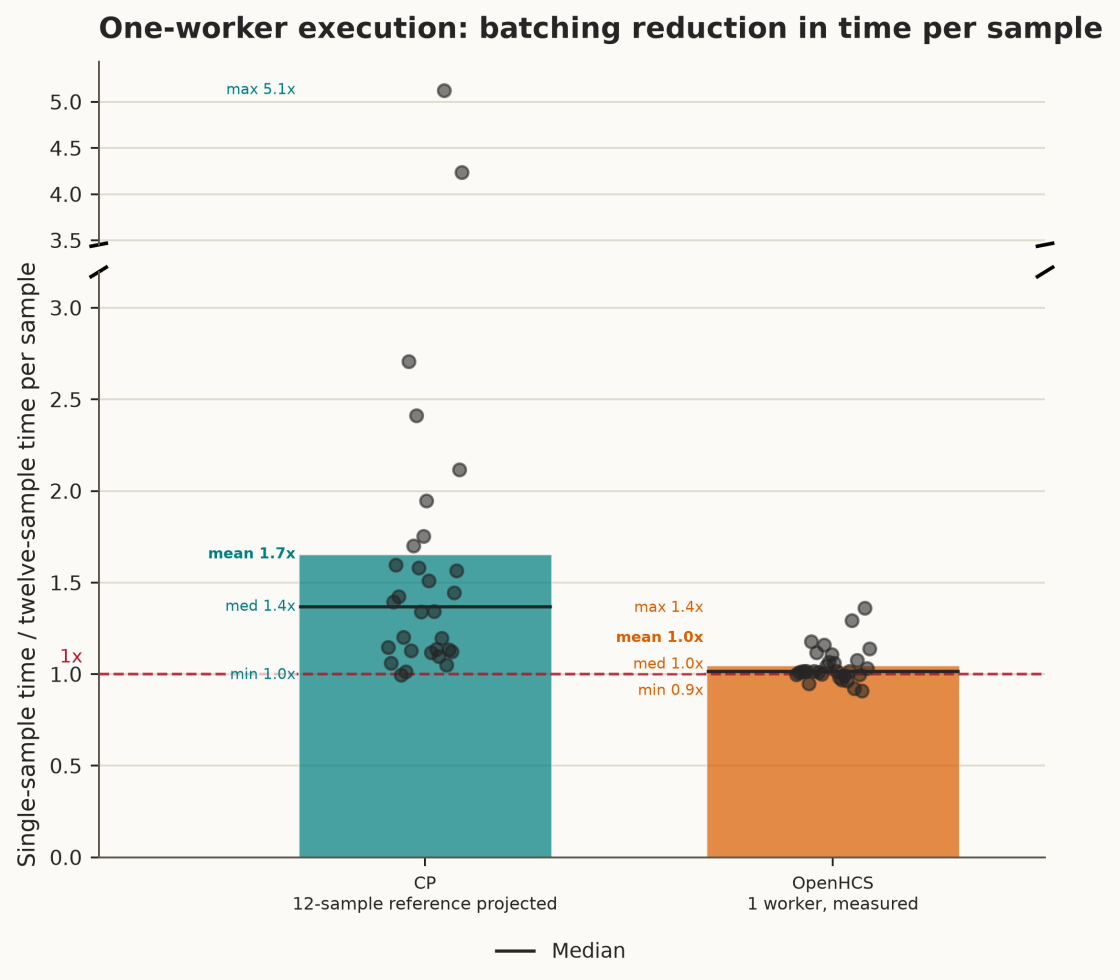
Supplementary Figure 6, continued. One-worker batch comparisons



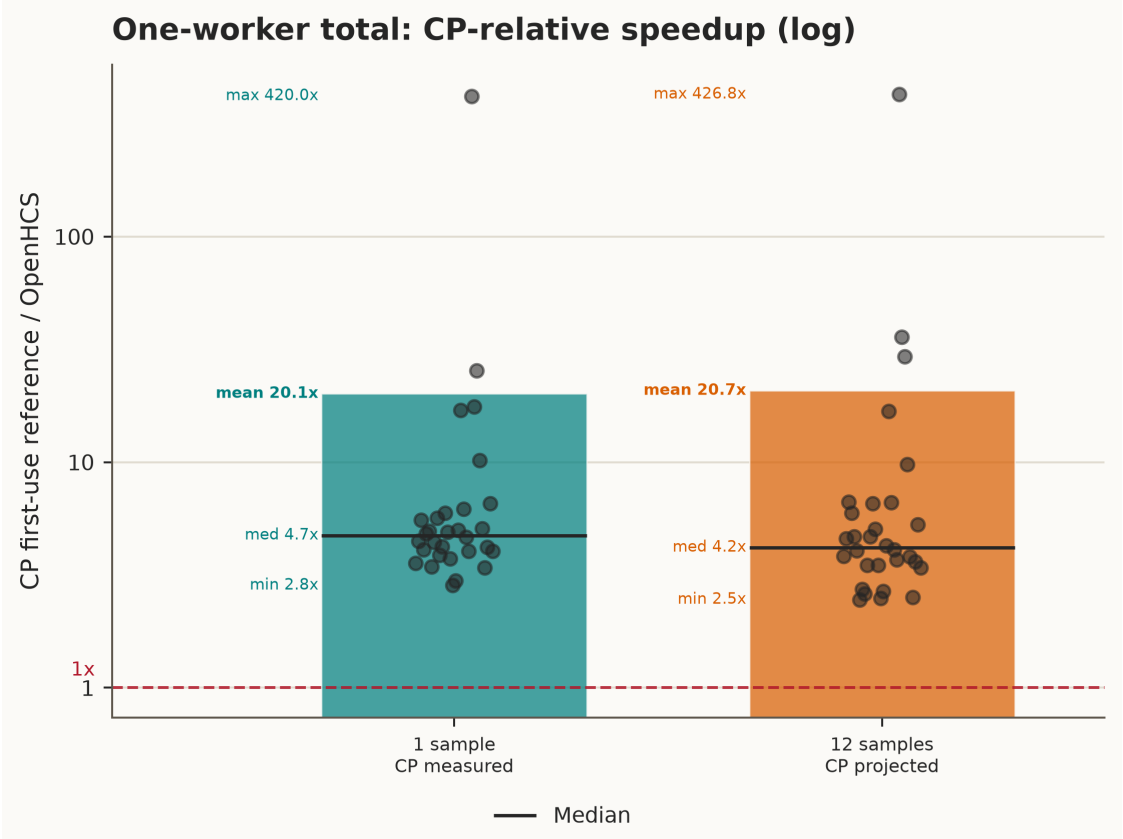
One-worker execution speedups for thirty workflows. CellProfiler's one-sample first batch is measured; its twelve-sample reference is projected from measured serial eight-sample first/warm batches. OpenHCS uses medians of three measured repetitions. Points are workflows, bars means and black lines medians. Supplementary Data 3 gives CPU affinities and projection calibration.



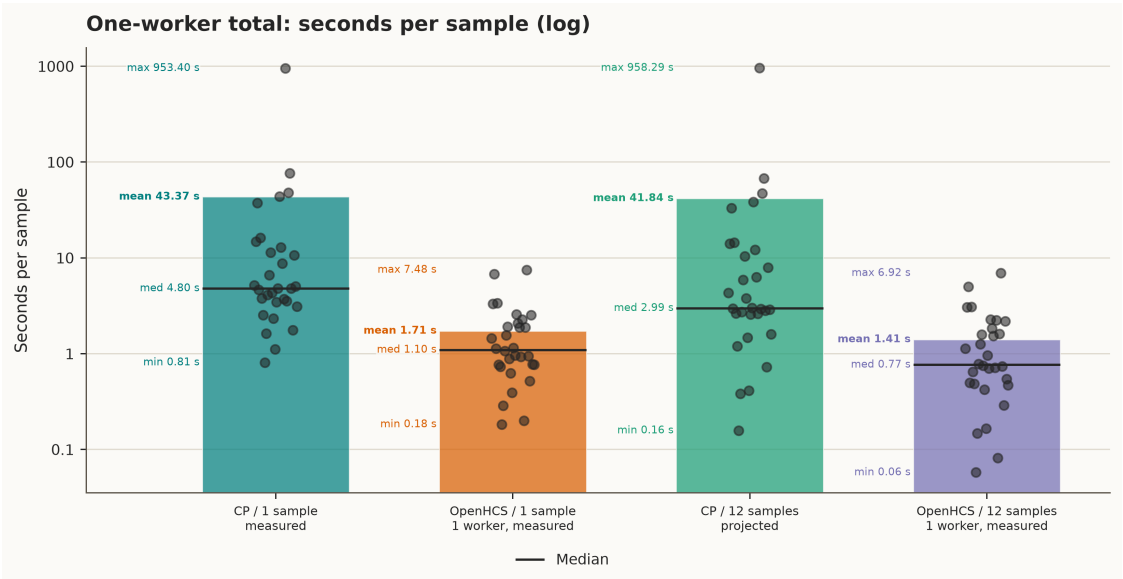
Execution seconds per sample for thirty workflows, dividing batch duration by sample count. CellProfiler's one-sample times are measured and twelve-sample times projected; OpenHCS times are measured medians of three repetitions. Both systems use one processing worker.



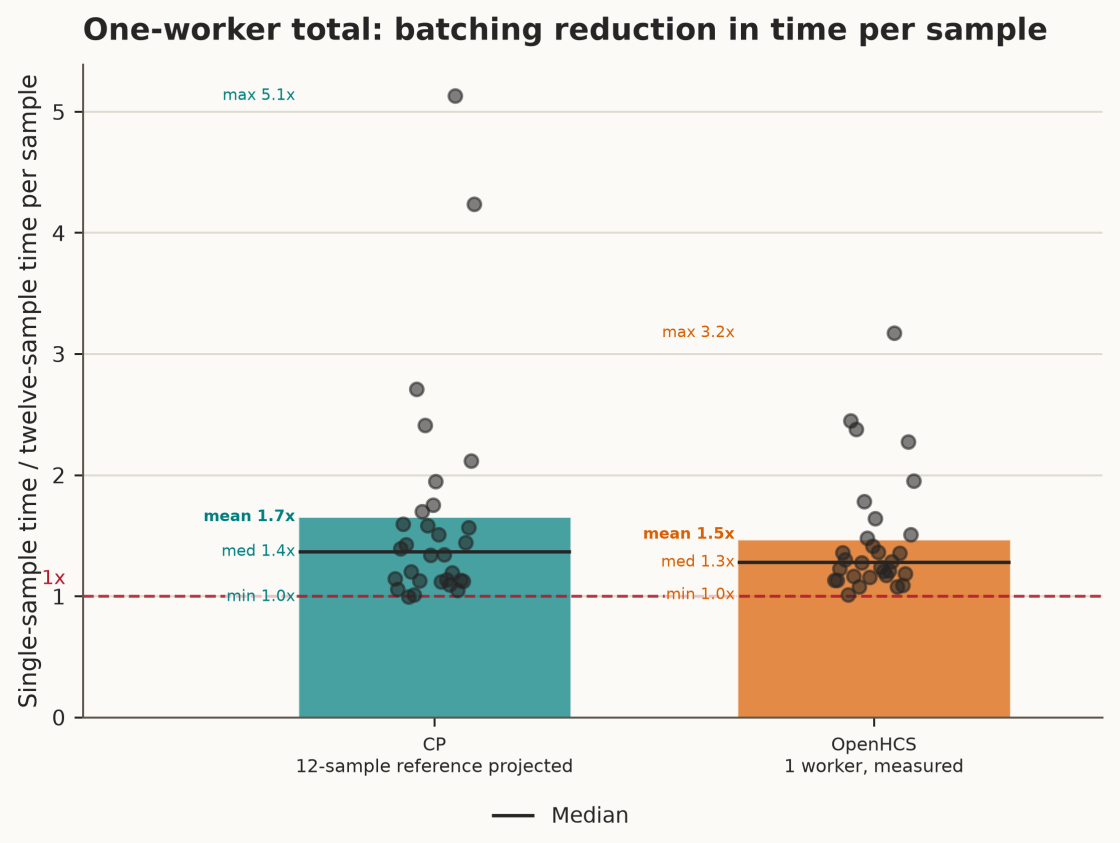
Reduction in execution time per sample across thirty workflows: single-sample time divided by twelve-sample time per sample. Values above one indicate lower time per sample. CellProfiler's twelve-sample times are projected; OpenHCS times are measured with one worker.



One-worker compile-plus-run speedups for thirty workflows. CellProfiler's one-sample first batch is measured and twelve-sample reference projected. OpenHCS uses medians of three measured repetitions, including client compilation. Points are workflows, bars means and black lines medians.



Compile-plus-run seconds per sample for thirty workflows, dividing total batch duration by sample count. CellProfiler's twelve-sample times are projected; OpenHCS times are measured medians of three repetitions with one worker and include client compilation.



Reduction in compile-plus-run time per sample across thirty workflows. Ratios compare a single sample with twelve-sample time per sample, including OpenHCS compilation. CellProfiler's twelve-sample times are projected; OpenHCS times are measured with one worker.

Supplementary Table 1. Reusable libraries and their roles

Library	Role in OpenHCS
metaclass-registry	Discovers classes implementing a shared interface and makes them available for selection.
python-introspect	Reads a function's parameters, types, defaults and documentation.
ObjectState	Tracks editable settings and resolves shared defaults and local overrides.
pyqt-reactive	Generates parameter controls and updates them as settings change.
pycodify	Generates editable Python representations and manages their imports.
ArrayBridge	Converts arrays between supported libraries and manages their computational resources.
PolyStore	Reads, writes and streams data through supported storage interfaces.

Library	Role in OpenHCS
ZMQRuntime	Coordinates communication, startup, shutdown and progress between processes.

Supplementary Table 2. Software ecosystems and project-hosted learning resources

Development history and extension interfaces. Release dates and first descriptions are distinguished below; they are not interchangeable measures of a project's age.

Platform	Release or description	Implementation and extension route
ImageJ / Fiji	ImageJ released in 1997; Fiji first released in 2007.	Java-based ImageJ supports plugins and macros; Fiji packages ImageJ with community plugins and an updater [1,2].
CellProfiler	Released in 2005; first described in 2006.	Configurable analysis modules; the later Python implementation also supports plugin modules [3–5].
Icy	Introduced in 2011.	Java plugins, scripting and graphical workflows [6,7].
napari	First minor release in 2019.	Python viewing and analysis; plugin packages add readers, writers and interactive tools [8,9].
PyImageJ	Described in 2022.	Connects Java-based ImageJ and Python within a Python process; napari-imagej uses this bridge within napari [10,11].
OMERO	Data-management system described in 2012.	Shared image storage, metadata management and access from analysis tools [12].
BioImageIT	Described in 2022.	Links image-data management with analysis tools and packaged processing environments [13].

Web inventories were inspected on 15 October 2026. Counts describe the named collections, not every resource available for a platform. A project-hosted resource can be community-contributed; hosting does not imply that every item is maintained or validated. Plugins, tutorial pages, script files and complete workflows are different units and should not be ranked as equivalent capabilities.

Platform	Extensions	Tutorials and training	Scripts and workflows
ImageJ / Fiji	>500 plugins reported in 2012 [1]; historical count.	26 pages under "Tutorials on this site" in the index.	402 top-level macro/script files in the archive.
CellProfiler	24 supported plugins; >90 built-in modules reported separately [5].	10 main tutorial exercises.	16 application collections, each potentially containing multiple pipelines.
Icy	448 plugins [14].	6 project-produced training PDFs, including a shortcut reference.	82 scripts; 87 graphical protocols.
napari	680 plugins in the dashboard [15].	4 worked tutorials in the index.	115 examples in the code gallery.
BioImageIT	Python tools with declared dependencies and environments; total not established.	2 worked tutorials in the documentation index.	No separate total established from the inspected index.

Counting rules. Tutorial counts exclude navigation links, translations of the same exercise, linked external courses, how-to sections and video playlists. The CellProfiler count runs from beginner segmentation through UnmixColors, excluding bonus material, profiling resources and older-version tutorials. ImageJ archive counts use unique top-level filenames ending in `.txt`, `.ijm` or `.js`, excluding `AA_ReadMe.txt`, subdirectories and ZIP bundles; a file can contain several macros. Napari examples are unique `gallery/*.html` destinations. Icy uses its catalogue totals and the six PDFs under "Our own course and tutorial," not six equivalent tutorials. An unestablished count is not zero.

Community support. Rueden et al. reported approximately 10,000 image.sc users and more than 15,000 posts per year in 2019 [16]. Ouyang et al.'s 2023 sample found installation mentioned in 9.9% of opening posts and approximately 19% of roughly 500 threads [17]. Napari's dashboard reported 1,460 tagged topics

and 938 participating users on 15 October 2026 [15]. Posts are individual messages; topics are discussion threads. The napari total does not distinguish bugs, usage questions, integration requests or announcements. Comparable per-platform bug/help counts were not established: direct image.sc access was unavailable, and neither the published sample nor project resource catalogues provide that breakdown. Missing counts must not be interpreted as absence of support requests, and raw forum volume is not a failure rate.

Supplementary Data 1. CellProfiler workflow comparison

Continuous integration

The versioned integration workflow defines the operating-system, Python-version, backend, microscope, numerical-parity and installed-desktop matrices. The Official30 comparator runs the manifest-owned workflows through ZeroMQ and compares their declared outputs against retained native CellProfiler values. These CI checks are separate from the timed performance captures.

Python 3.11 and 3.13 each run the generated CellProfiler workflow corpus through multiprocessing and the ZeroMQ server on Linux, Windows and macOS. Python 3.12 tests every combination of disk or Zarr storage and ImageXpress or OperaPhenix inputs on those systems. A separate Python 3.14 Linux job checks the installed core runtime without the optional Centrosome comparison dependency. The full thirty-workflow numerical comparison runs through ZeroMQ against committed native CellProfiler references on Linux with Python 3.12 and on all three operating systems with Python 3.14.

The full integration matrix runs nightly, on release tags and on demand; PyPI publication requires a successful run on the exact release commit. Pull requests run focused checks and structural guardrails rather than the full matrix. Additional jobs exercise package installation, the GUI and installed desktop candidates through MCP, including Intel and Apple Silicon macOS, and native Windows and macOS installers.

Import, export and comparison methods

Imported `ExportToDatabase` modules run once per plate after image-group processing. They collect the selected images, objects, measurements, relationships, thumbnails and grouping information into CellProfiler Analyst tables [18]. The export produces a self-contained SQLite database and matching `.properties` files. Non-SQLite databases, custom filter rows, `.workspace`

generation and some historical aggregation settings remain unsupported; unsupported requests fail or are identified in the compatibility documentation.

The OpenHCS 0.8.5 release test preceded the unified 30-workflow value comparison below.

An earlier matched performance evaluation compared the declared table, database and image outputs for all 30 workflows in a warmup and three measured repetitions per engine. All 120 OpenHCS observations matched complete native CellProfiler runs. The original matched benchmark gives its output inventories and timing boundaries. Supplementary Data 3 describes the subsequent first-use-inclusive worker sweep used in main Figure 6.

For the five workflows without file exports, terminal image or object-label exports were appended while preserving the original processing modules and settings. Native CellProfiler generated eight additional reference artifacts. A subsequent unified run compiled and executed all 30 workflows afresh and compared each candidate with its selected native reference values. Object labels were compared exactly after singleton-axis normalization; numerical images used the stated float tolerances. The following sections give the export definitions, reference inventory and per-workflow comparisons. The original unified-run environment was not recorded; the five-workflow export audit has its own recorded environment.

The corpus contains 22 workflows and associated image sets from the official CellProfiler 3 examples repository, seven workflows and image sets from the official CellProfiler tutorials repository, and one workflow from the supplement to the CellProfiler 4 performance study [19–21]. The CellProfiler project and the cited dataset contributors retain authorship and provenance for these materials. The retained manifest maps workflow names to pipeline and image locations; Supplementary Data 1-3 provide the corresponding comparison, coverage and throughput tables.

This supplement indexes source tables and evaluation records. Figure scripts derive panels and plotted-row exports from the linked files; paths are relative to this file.

The acquisition scripts pin the official CellProfiler examples to `4972b59e670a4ae96c3d453803c92eeff378d054`, the official tutorials to `264a8155da21a2d468051f78211bed2e580a8934`, and the CellProfiler 4 benchmark supplement to `40abc2e600fd46b74c213999dd25c5245048dc92`.

Declared exported-file coverage

The matched run checked every declared exported file in each workflow: a file-level fraction of 1.00. Counts below come from the original matched-run reports, with one warmup and three measured repetitions. All four observations per workflow had complete output inventories and zero image, CSV or database differences. The original comparator rejects files without a value-comparison route. This fraction describes exported files; internal intermediates and individual measurement columns are not its denominator.

Workflow	Declared files checked / declared files
ExampleColocalization	1 / 1
ExampleCometAssay	6 / 6
ExampleFly	7 / 7
ExampleFlyURL	7 / 7
ExampleHuman	7 / 7
ExampleIlluminationCorrection_Example1_AllMethod	1 / 1
ExampleIlluminationCorrection_Example1_EachMethod	1 / 1
ExampleIlluminationCorrection_Example2	3 / 3
ExampleIlluminationCorrection_Example3	2 / 2
ExampleImagingFlowCytometryObjectsInGrid	1 / 1
ExampleNeighbors	4 / 4
ExamplePercentPositive	5 / 5
ExampleSpeckles	3 / 3
ExampleTrackObjects	23 / 23
ExampleTumor	3 / 3
ExampleUntangleAndStraightenWorms	1 / 1
ExampleUntangleWorms	4 / 4
ExampleUntangleWormsBrightField	1 / 1
ExampleVitra	5 / 5
ExampleWoundHealing	1 / 1
ExampleYeastColonies	3 / 3
ExampleYeastPatches	7 / 7
cp4_supplement_combine_objects	1 / 1
cp_tutorial_3d_monolayer	8 / 8
cp_tutorial_advanced_segmentation_final	8 / 8

Workflow	Declared files checked / declared files
cp_tutorial_beginner_segmentation_final	9 / 9
cp_tutorial_pixel_based_classification	3 / 3
cp_tutorial_quality_control	3 / 3
cp_tutorial_translocation_final	3 / 3
cp_tutorial_translocation_start	1 / 1

Unified 30-workflow value comparison

The unified evidence directory preserves one immutable run in which all 30 workflows were compiled and executed afresh through isolated ZMQ execution endpoints. All 30 selected reference-value comparisons were equivalent and reported zero differences. Its fresh reference inventory contains 21 CSV profiles, three SQLite profiles and six image- or array-only profiles. Image comparison executed for seven workflows, including the completed translocation overlay compared alongside its SQLite values.

The observations, phase timings, and summary retain the comparison results. The original observations identify the candidate as OpenHCS 0.8.5 and retain endpoint, executable and submitted-pipeline provenance. The original run environment was not recorded in this bundle; versions from later runs cannot supply it.

Five-workflow image and object-label exports

Five source workflows compute images or objects without exporting files. The export-extension manifest selects versions of those workflows with terminal exports added. Its generated pipelines retain the original processing modules and settings. Native CellProfiler produced the exported references; OpenHCS processed the same scoped inputs.

Workflow	Exported values compared	Result
Combine objects	Combined-object labels	Exact label agreement
Translocation starter	Nuclear labels	Exact label agreement
Illumination Example 1, EachMethod	Corrected green image	Float-tolerance agreement
Illumination Example 2	Uncorrected, small-block-corrected and large-block-corrected nuclear labels	Exact agreement for all three label images
Illumination Example 3	Polynomial-corrected and convex-hull-corrected images	Float-tolerance agreement for both images

The per-artifact comparison table records eight passing artifacts. All five integer label images agree exactly after normalization of singleton dimensions. The three numerical images have zero pixels outside the absolute and relative tolerances of `1e-6`; their maximum absolute difference is `2.9802322387695312e-8`. The per-workflow observations link these comparisons to fresh candidate executions and the retained native outputs.

The run environment identifies OpenHCS execution source `7ca8ecb8e73a882ef0f15616d57b00f0dabf73e0`, Python 3.12.3, NumPy 2.1.3 and SciPy 1.18.0. Native references were generated with CellProfiler 4.2.8.1 on Python 3.9.25, NumPy 1.24.4 and SciPy 1.9.0. Each candidate ran through a fresh matching OpenHCS endpoint. Source revisions, native-reference origin and endpoint identities are retained with the results. These exported references are incorporated into the unified 30-workflow run; the earlier five-workflow audit and release records retain their own protocols and identities.

Supplementary Data 2. Archived CellProfiler coverage

- Module names and associated workflows.
- Individual settings and recorded handling categories.
- Processing registration and corpus membership.

These files describe the archived coverage report. Setting categories distinguish bound parameters, artifact contracts, infrastructure, and intentionally ignored settings. Registration or a handling category is not itself an output-equivalence test. Module aliases and source/export roles differ between the tables, so their category totals must not be added as mutually exclusive classes. They are not a current-version compatibility matrix.

Supplementary Data 3. Worker measurements

Single-sample timing summaries

With one sample, one worker and one numerical thread, execution was a median 6.965-fold faster than native CellProfiler, with minimum 3.886-fold. Compile-plus-run speedup was a median 4.740-fold, with minimum 2.844-fold. Main Figure 6B shows all thirty workflows. Native times include internal initialization within the first pipeline call; external process/server startup and OpenHCS library readiness and warmup are excluded. OpenHCS times are medians of three measured repetitions after warmup. Compile-plus-run time includes client compilation plus execution submission/wait; nested server and worker durations are counted once.

Matched-core CellProfiler projection

Main Figure 6C estimates parallel CellProfiler execution from measured serial batches of eight repeated samples, dividing the workload evenly among independent single-threaded processes. For N total samples and C processes, projected execution time is `T8_first + (N/C - 8) * median(T8_warm)/8`. Each process retains its initialization term; total time adds the first batch's pre-pipeline overhead once, not once per sample. Concurrent processes are assumed equally fast, with no contention or coordination overhead. These estimates assume parallel execution; multicore CellProfiler runs were not measured. The measured eight-sample batch used one numerical thread with access to four CPU cores; the projection assumes the same processing rate on one core per process.

The retained serial calibration tested three workflows at 8, 12 and 16 samples. A model based on a two-sample batch had maximum absolute error 5.21%; the model based on an eight-sample batch from the same measurement session, used here, had maximum error 7.06%. Using an eight-sample batch from an earlier session underpredicted later observations by up to 14.03%. These errors are not multicore confidence intervals. Per-process counts of 3, 4 and 6 in Figure 6C extrapolate below the validated batch sizes; actual matched multicore measurements are still needed. The one-worker/twelve-sample column stays within the serial model's validation range, though validation covered only three workflows.

Individual projected and measured times and source identities and assumptions retain every plotted value. The source reports preserve measured times separately from estimates. Figure 6A summarizes the module tests; the per-module results and test sources are linked below.

Serial batching and preparation amortization

The one-worker controls compare one and twelve samples. Both systems amortize preparation across a batch: CellProfiler initialization remains inside its first execution, while OpenHCS library readiness precedes pipeline timing and compilation is included in total time. The twelve-sample CellProfiler times are estimated from measured serial eight-sample batches before and after initialization; twelve-sample times were not measured in this thirty-workflow comparison. Each panel retains all thirty workflows. Different CPU affinities between the single-sample and batch captures are retained in the record; these plots show the observed comparison rather than isolating initialization alone.

- One-worker CP-relative speedups: execution and compile-plus-run total.

- Time per sample: execution and total.
- Batching reduction in time per sample, relative to one sample: execution and total.

Thirty-workflow timing measurements

- Benchmark protocol and timing definitions, with the seven-configuration protocol.
- Single-sample execution and total, linear and logarithmic, with manuscript numerical claims.
- Charts and source tables: per-workflow speedups, worker summary, assignment summary, output agreement, and graded module coverage, with the per-module behavior and test attribution.
- Variable-batch processing speedups: linear, logarithmic, and all paired clocks and ratios.
- Variable-batch setup-plus-processing speedups: linear, logarithmic, and all paired clocks and ratios.
- Measured processing times for twelve repeated samples: linear, logarithmic, scaling table, and parallel efficiency.
- Measured setup-plus-processing times for twelve repeated samples: linear, logarithmic, scaling table, and parallel efficiency.
- Measured one- and eight-sample CellProfiler batches, calibration table, and projection validation.

All thirty workflows met the output-comparison criteria. One- and eight-sample CellProfiler times were measured; twelve- and sixteen-sample times were estimated. Every workflow had a processing and setup-plus-processing speedup above one in each measured or projected comparison, including the one-worker, twelve-sample batch comparison. With twelve repeated samples held fixed, every workflow also ran faster with two, three and four OpenHCS workers than with one. These OpenHCS scaling results use measured times throughout.

The one- and eight-sample CellProfiler measurements permitted different CPU counts. Checks of the timing model covered three workflows, rather than the full set of thirty. The linked record reports prediction errors within and between measurement sessions. Combined memory use across processes was not measured.

Historical timing panels and scaling protocols are in the archived supplementary source linked above.

Figure assembly and interface records

The supplement generator combines existing artwork and saves editable SVG, PDF and PNG outputs. It changes layout only; measurements, segmentation and display contrast are unchanged.

Source illustrations show array grouping and scheduling, function preparation, connected images and measurements, custom-function registration, CellProfiler translation, and Fiji/napari inspection.

The native code/field interaction record documents the parameter-edit round trip. Main Figure 1 uses the two-plate capture, with ExampleHuman's nine-step recipe displayed, and the native custom-function form and code capture. The capture-source record identifies the source state at capture. The custom function has gain 1.2 and offset 0.0; the code readback identifies the same function. These captures show authoring controls; the function was not executed on an assay.

The translation source maps ExampleCometAssay to 12 OpenHCS steps and checks the generated Python round trip. Image and ROI inspection examples are attributed in the gallery source record.

Supplementary Data 4. Method-directed NeuronCyto demonstration

A gpt-5.6-sol agent analysed NeuronCyto II image 1 through MCP using a method-directed brief. The brief specified channel roles, normalization, neurite enhancement, nucleus and soma segmentation, rooted neurite assignment, and morphology outputs. The original run produced nine neurons, ten nuclei, eight branches, 25 graph paths and 1,982 pixels of total outgrowth. It demonstrates authoring, execution and inspection; no manual segmentation score was supplied.

The original evaluation contains the prompt, input hashes, software/model versions, tool calls, outputs and viewer checks. The figure source identifies the original recording. Later corrections and replays used the same images with their own pipelines and outputs.

Figure artwork uses unmodified Lucide icons under the licence; the source record identifies their revision.

Separate evaluation after neurite algorithm changes

A separate OpenHCS 0.8.5 replay used the same two public images with nuclear-supported soma detection and soma-rooted path assignment. It produced eight neuronal cell bodies, eight nuclei, 18 processes, two branch events, one resolved crossover and 24 graph paths. The eight per-cell table totals agree with the graph distance features and sum to 2556.137 pixels under unit spacing. The evaluation summary identifies its execution, saved source and outputs separately from the original demonstration and the OpenHCS 0.7.14 correction.

Published manual-tracing references

The NeuronCyto II publication supplies independent measurements for image 1 in its supplementary files 8–10. The two input TIFFs used here are byte-identical to `CrossOvers_Images/1_w1.tif` and `1_w2.tif` in the public `Testing image.zip` archive. The checksums are listed in the original run record.

Published reference	Image 1 entry	Source file
Manually traced neurons	Eight per-cell entries	Supplement 9
Neurite length by branch order	Primary 3,633.672; secondary 198.929; tertiary 0; total 3,832.601	Supplement 8
Crossover count	Three crossovers; NeuronCyto II resolved two	Supplement 10

Lengths above retain the numerical units of the source tables. The eight per-cell reference lengths are 426.82, 172.375, 751.962, 395.04, 669.783, 343.629, 587.841 and 485.152. These entries give a reference against which to investigate the nine reported neurons. Spatial cell matching, measurement-unit alignment and agreement on the definition of neurite length are needed before computing per-cell accuracy. The public testing archive contains the images; downloadable manual trace coordinates have not been located. The published crossover count and the OpenHCS branch count measure different quantities.

Main Figure 4C–D and F–H are full-resolution presentations of the frozen raw arrays, body-label planes and graph-path archives, rather than reduced native viewer screenshots. The source manifest records the six original file hashes and the recorded raw display windows. Dark label colours and vector strokes are display choices only; no analysis was rerun or coordinates changed. Figure 4E uses an unmodified embedded JPEG from page 4 of the published article PDF, giving a 450-pixel panel crop instead of the earlier 225-pixel web crop. Its source, extraction and attribution are recorded in the published-reference record.

The reference audit records the source links, file checksums and ROI identifiers. These measurements were examined after the recorded agent run and were not supplied to the agent during authoring.

Supplementary Data 5. Complex CellProfiler workflow imports

The derived workflow report lists every imported step, its function names and call count for the advanced segmentation and 3D monolayer tutorials. It links the source pipelines and generated editable Python. The generator checks function identities and parameters after reloading each document and records the hashes of the source pipelines, importer and function implementations.

These are import and code-round-trip checks. Supplementary Data 1 separately records the release CI executions and selected output comparisons for both workflows.

Supplementary Data 6. Workflow regression tests

The tests exercise representative authoring and validation cases alongside the recorded UI demonstration. The following source files are pinned to commit `7a7d21fee726905b87b9020cebf0bdfc350631f4`; these three test files are unchanged from the OpenHCS 0.8.5 release commit.

Tested behavior	Source tests
Nested and inherited configuration, parameter order, enum identity and function-step reconstruction	Python generation tests
Rejection of incompatible memory types, grouping, required axes and stack configuration	Compiled-function validation tests
Rejection of a function-detail request using a stale catalog revision	Function-catalog tests

The unit-test job passed while selecting `tests/unit` and `tests/core`. These focused tests include controlled fixtures and isolate the stated behavior. The real-corpus execution and value comparisons are provided by the separate Official30 job in Supplementary Data 1.

Supplementary Data 7. Prospective agent-authored assay validation

Nuclear detection 1, cell-boundary 1 and protein translocation 1

These three analyses were applied without changing their processing settings after their held-out images were disclosed (Supplementary Figures 2–3). Nuclear detection analysis 1 matched 4,733 of 5,720 reference nuclei on 50 BBBC039 fields at intersection over union at least 0.5. Pooled precision was 0.680, recall was 0.827 and F1 was 0.746; mean field Dice area overlap was 0.935. Precision measures how many detections are correct, recall how many reference nuclei were found, and F1 balances both; 1 is best. The workflow predicted 6,964 nuclei, 1,244 more than the reference, and more reference nuclei were split than merged. It therefore recovered nuclear area well while often splitting nuclei into multiple detections.

Cell-boundary analysis 1 placed 29,450 of 43,875 predicted adjacent-cell boundary pixels within two pixels of a manual outline on 12 BBBC007 fields, a pooled fraction of 0.671. The workflow predicted 1,274 nuclei and 1,273 cells; every predicted nucleus overlapped a cell and one cell shared two nuclei. The manual outlines enclosed 1,082 closed nuclear interiors, while 12 open or frame-connected regions were excluded. Because the score checks predicted borders against manual outlines, it can reward an incomplete segmentation and does not establish that each detected cell matches a reference cell.

Protein translocation analysis 1 produced matched nuclear and cytoplasmic measurements for 14,262 cells across all 92 held-out BBBC013 wells. Wortmannin controls separated with Z′ of 0.751 and mean nuclear-to-cytoplasmic GFP ratios of 7.235 and 0.915 for positive and negative controls. LY294002 controls gave Z′ of 0.554 and means of 7.219 and 1.127. Z′ measures control separation relative to variation between wells; both scores exceed the 0.5 threshold for an excellent screening assay. Both held-out dose series showed the expected increase in nuclear translocation. This result evaluates recovery of the assay response; BBBC013 does not supply manual masks with which to score segmentation.

The validation report gives the prospective design and results for BBBC039, BBBC007 and BBBC013. Each trial used a fresh gpt-5.6-sol author through the connected OpenHCS desktop and MCP surface. Four development fields or wells were visible before the scientific pipeline was frozen; held-out references and treatment metadata were then scored by the typed evaluator in `benchmark/annotated_validation.py`.

The evidence index links final pipelines and held-out scores. The preparation record describes deterministic partitions, image normalization and evaluation definitions. Prediction arrays, source images and file inventories are indexed in the linked archive records.

The prospective held-out panels in Supplementary Figures 2–3 are generated by `build_slas_independent_validation.py` from the three held-out scores. Its plot data and source record link the plotted values to the original evaluations.

Supplementary Data 8. Autonomous analysis evidence

Evaluation endpoints and reference types

Assay	Endpoint	Reference
H001 bright objects	One-to-one object F1	Computational notebook labels
BBBC039 nuclei	One-to-one object F1 and foreground Dice	Independent instance annotations
BBBC007 DNA/actin	Directed boundary fraction	Manual-outline union
BBBC013 translocation	Well-level GFP ratio and control Z′	Treatment groups
H002 3D centres	Matched-centre voxel distance	Manual centre annotations
Retinal somata	Counts and distributed image review	Matched raw/result views
Public and laboratory neurites	Shaft and path recovery	Matched raw/path views

Trial-level autonomous results

Analysis names describe the task; numbers distinguish independently developed methods. A correction remains part of the same numbered analysis. Multiple measurements from one analysis are listed separately. The linked trial records map these names to original files and identifiers.

Assay	Model	Data partition	Endpoint	Result
Nuclear detection 1 (BBBC039)	gpt-5.6-sol	50 held-out fields	Object F1; foreground Dice	0.746; 0.935
Nuclear detection 2 (BBBC039)	gpt-6.1-sol	175 uninspected; all 200 fields	Pooled object F1, respectively	0.910; 0.906

Assay	Model	Data partition	Endpoint	Result
Nuclear detection 3 (BBBC039)	gpt-6.1-sol	175 uninspected; all 200 fields	Pooled object F1, respectively	0.906; 0.898
Nuclear detection 4 (BBBC039)	gpt-6.1-sol	175 uninspected; all 200 fields	Pooled object F1, respectively	0.906; 0.904
Nuclear detection 2, repair	gpt-6.1-sol	Same three development fields	Object F1, first to final	0.908 to 0.934
Cell-boundary analysis 1 (BBBC007)	gpt-5.6-sol	12 held-out fields	Directed boundary fraction	0.671
Cell-boundary analysis 2 (BBBC007)	gpt-6.1-sol	16 fields including development	Directed boundary fraction	0.743
Cell-boundary analysis 3 (BBBC007)	gpt-6.1-sol	16 fields including development	Directed boundary fraction	0.740
Paired nuclear/cell detection (H003)	gpt-6.1-sol	One development field	Nuclei; actin-supported cells	56; 54
Protein translocation 1 (BBBC013)	gpt-5.6-sol	92 held-out wells	Control Z′, Wortmannin; LY294002	0.751; 0.554
Protein translocation 2 (BBBC013)	gpt-6.1-sol	96 wells including four development wells	Control Z′, LY294002; Wortmannin	0.849; 0.726
Bright-object detection 1 (H001)	gpt-6.1-sol	One development image	Computational-reference F1, first to final	0.929 to 0.944
Bright-object detection 2 (H001)	gpt-6.1-sol	One development image	Final labels; matched image review	62; no reference score
3D nuclear localisation 1 (H002)	gpt-6.1-sol	One development volume	Centres within 30 voxels; mean matched error	15 of 15; 4.80 voxels
3D nuclear localisation 2 (H002)	gpt-6.1-sol	One development volume	Centres within 30 voxels; mean matched error	15 of 15; 4.86 voxels

Assay	Model	Data partition	Endpoint	Result
Retinal cell detection 1	gpt-6.1-sol	One development field	Detected objects; image review	102
Retinal cell detection 2	gpt-6.1-sol	One development field	Candidates; border candidates	136; 10
Public-image neurite tracing	gpt-6.1-sol	One development field	Principal-shaft recovery	Matched image review
Laboratory-image neurite tracing	gpt-6.1-sol	Nine development fields	Soma and path recovery	Matched image review
Neurite drug-response analyses 1–3	gpt-6.1-sol	180 fields each; six/four/eight development fields	Control-normalized outgrowth response	Both drug responses recovered by all three; magnitudes varied

Additional assays: detailed results

The main text groups these single- and two-agent demonstrations by the corrections and measurements they demonstrate. Their detailed results are given here; the trial-level table above lists every score.

Cell-boundary analyses (BBBC007). Analyses 2 and 3 reached directed boundary fractions of 0.743 and 0.740 across 16 DNA/actin fields; analysis 1 reached 0.671 across 12 held-out fields (main Figure 5C; Supplementary Data 7). In one paired-field analysis, the agent detected 56 nuclei and kept 54 actin-supported cells after removing two seed-only candidates. Each cell mask included all pixels of its associated nucleus. Main Figure 5F shows a same-region merge repair.

Bright-object detection. Analysis 1 matched 59 of 64 computational-reference objects and raised F1 from 0.929 to 0.944 through self-directed repair (main Figures 3B and 5A). The matched views show correction of an elongated object's split and recovery of a pair lost during an intermediate revision.

3D nuclear localisation. Analysis 1 recovered all 15 annotated 3D centres within 30 voxels, with mean matched error of 4.80 voxels (main Figure 5D,I); analysis 2 also recovered 15 of 15, with mean matched error of 4.86 voxels. Eleven predictions of the first agent had no matching annotation; because annotations cover only some centres, these are not yet classified as false detections. A

different agent repaired an internal body partition at Z=36 while keeping its neighbour separate (main Figure 5H).

Retinal cell detection. Analysis 1 detected 102 objects after improving fragmented soma footprints while keeping a neighbouring pair separate (main Figure 5G). Analysis 2 detected 136 candidates, including ten at the border. Distributed raw/result review covered bright somata, diffuse rings and crowded regions (Supplementary Figure 4).

Neurite fields. One autonomous agent analysed all nine laboratory neurite fields and revised its tracing sensitivity after inspecting thin processes (main Figure 4F–H). Matched sparse and dense views showed recovery of raw-visible paths with the same soma labels in the reviewed field. In the public NeuronCyto II field, the initial pipeline recovered principal shafts and junctions (main Figure 4C–E); the final repair extended beyond that target. The initial result is shown alongside the published algorithm output.

BBBC039 field-level errors and repair

The per-field scores for the 175 fields that none of the three agents opened (rows with `opened_by_any_of_three_authors=False`) give the following pattern. Pooled counts reproduce the reported F1 values of 0.910, 0.906 and 0.906.

Agent	Fields with F1 ≥ 0.90	Fields below 0.80	Median field F1	Missed nuclei	False detections	Fields with more misses than false detections
Nuclear detection 2	121	6	0.919	2,661	951	171
Nuclear detection 3	117	7	0.917	2,929	806	173
Nuclear detection 4	121	6	0.915	2,743	1,033	171

Misses made up 73% to 78% of each agent's errors. The same six fields (20641_A12_7, 20591_L01_3, 20590_D08_7, 20586_O15_6, 20626_E05_2 and 20630_P09_4) scored below 0.80 for all three agents; nuclear detection analysis 3 also scored 0.727 on 20589_P23_7. These six fields hold 17 to 180 reference nuclei (median 59), against a median of 126 across all 175 fields. In field 20641_A12_7, all

three agents missed 57 to 59 of 69 nuclei, giving F1 of 0.23 to 0.27. Field F1 was only weakly related to the number of nuclei (Spearman rank correlation ρ −0.15 to −0.24, indicating a weak tendency for more crowded fields to score lower); in the third of fields with the most nuclei (138 to 199), pooled F1 was 0.893 to 0.897, against 0.920 to 0.928 in the middle third.

A same-input repair in nuclear detection analysis 2 lowered the maxima-suppression size from 14 to 10 and raised F1 from 0.908 to 0.934 on the three inspected fields, recovering 14 missed nuclei while split reference nuclei increased from 28 to 43. The repair was not separately evaluated on the full collection. Between two independent agents' final pipelines on all 200 fields, per-field F1 differed mostly by less than 0.03, and the lower pooled score came chiefly from 314 additional missed nuclei (Supplementary Figure 3).

Instructions used in the reported trials

Nuclear detection analyses 2–4 received different revisions of the OpenHCS guide. The three neurite drug-response analyses received the same revision. The trial resource table and treatment-assay record identify the delivered instructions by file checksum, so the exact versions can be recovered. No instruction-guide version was recorded for nuclear detection 1, cell-boundary 1 or protein translocation 1.

Source artwork and assisted mosaic review

Main Figure 4 shows public shafts, autonomous laboratory-field analysis and three independently authored treatment responses. The earlier assisted response comparison remains in Supplementary Figure 5. The separate assisted nine-field mosaic artwork and overlap/core layout show stitching at the acquisition coordinates. One pooled percentile pair was fitted per complete nine-field channel stack before assembly. Some faint segments were missed, and ownership was unresolved in crowded regions. The completed all-channel continuation reported 1,740 soma candidates and 123,054 micrometres of computed outgrowth at the declared spacing. These computed totals describe the continuation; the earlier panels show sampled regions. Overlapping fields and uncertain ownership prevent their interpretation as a unique biological census, and no spatial reference validates the lengths.

The independent BBBC039 repeat matched 20,207 of 23,615 reference instances, with 1,164 excess predictions and 3,408 misses. The earlier author matched 20,521, with 1,153 excess predictions and 3,094 misses. Pooled F1 values are 0.898 and 0.906, respectively, at intersection over union ≥ 0.5. The independent-author

sheet shows this comparison; annotation-empty fields coincide at the scatter origin.

The original artwork shows wider image regions and additional comparisons: process architecture, runtime composition, compiler preparation, typed extension, full-plate translocation, translocation development, H001 scored review, volume repair, orthogonal localisation, retinal review, retinal development, paired DNA repair, actin support checks, public shaft/junction review and public main shafts. Retinal source: user-provided R0010 RBPMS images; physical calibration unverified. DNA/actin source: BBBC007v1 A02, Sabatini laboratory, Whitehead Institute; CC0.

Scientific instructions supplied to authors

The instruction archive links each trial to its original brief. The table summarizes the scientific instructions; original trial identifiers and operational instructions remain in the archive. The thick-shaft-only target was clarified after the neurite run and was absent from its original brief.

Brief	Scientific instruction
1: BBBC007 nuclei and cells	Segment nuclei and seeded cells from paired DNA/actin images across 16 source sets, with labels and object tables. Source bindings and optional custom-function tracks were supplied.
2: Paired DNA and actin: full released sixteen-source collection	Analyse all sixteen paired DNA/actin fields, choosing the method from image measurements and packaged guidance. Inspect separated regions before selecting the final pipeline.
3: BBBC013 GFP translocation	Measure nuclear and cytoplasmic GFP in 96 paired source sets and produce cell/well response tables. Source bindings, catalogue tracks and optional statistics extensions were supplied.
4: BBBC039 nuclei	Segment nuclear instances in 200 source sets and save label images and object measurements. The brief supplied source bindings and catalogue-only authoring.
5: H001: bright-object segmentation	Segment bright objects in one two-dimensional image and report pixel areas and count. Review raw images and labels across positions and intensity windows.

Brief	Scientific instruction
6: H002: 3-D centre detection	Detect nuclear centres in a three-dimensional volume and report z/y/x voxel coordinates and count. Inspect separated slices and orthogonal views; physical spacing was unverified.
7: Paired nucleus and cell segmentation	Segment nuclei and actin-supported cells in one paired field, preserving w1/w2 identity. Explicit source pairing and distributed multi-window review were required.
8: H004 blind neurite-outgrowth analysis	Infer channel roles in one paired neurite field and measure supported somata and outgrowth. Review faint paths, background connections and crossing ownership before evaluation.
9: Paired-field soma and neurite-outgrowth analysis	Analyse paired soma/neurite images, infer channel roles and save object/path measurements in pixel units. Keep the first attempt and later self-directed repairs.
10: Blind neurite-outgrowth development task	Analyse nine paired fields from one coded well, with w1 DAPI, w2 FITC and declared spacing 1.3556 µm per pixel. Choose the method independently and report uncertain crossing ownership.
11: Personal nine-field neurite analysis	Analyse nine paired DAPI/calcein fields and their acquisition layout, including authorized mosaic work. Report fieldwise geometry without counting overlapping fields as unique cells.
12: Personal nine-field neurite analysis: retained development	Continue a saved nine-field analysis and assembled mosaics, reviewing misses, bridges and ownership. This is a retained-development continuation.
13: Unavailable pooled-stack scientific brief	The brief for this trial was not kept.
14: R0010 independent-author development trial	Segment RBPMS-positive retinal somata in one development CZI and report labels, measurements and count. RBPMS/Hoechst hints and distributed review were supplied. Retinal analyses 1–2 used fresh sessions; the archive documents their launch context.
15: Retinal whole-mount RBPMS analysis	Segment RBPMS-positive somata in the retinal development field, confirming channels from raw images. Preserve the first attempt, self-directed repairs and matched regional review.

Brief	Scientific instruction
16: Retinal whole-mount RBPMS analysis	Analyse the authorized retinal development input with RBPMS/Hoechst hints and packaged MCP guidance. Report uncertain soma identity, extent and boundaries.

The original instruction files for nuclear detection 1, cell-boundary 1 and protein translocation 1 were not retained; Supplementary Data 7 describes their design and partitions. Although an archived retinal brief mentions inherited conversation, the launch records for retinal analyses 1–2 show new sessions with memory disabled and no parent conversation. Other trials associated with that brief have not been checked for inherited context. The trial resource table links the original launch records.

The current skill describes inspection and repair practice, including additions made after earlier trials. Main Figure 3 shows that procedure alongside the H001 author's actual decisions. Wall times come from the trial resource catalogue, and the candidate sequence comes from the author's report; screenshot counts were not used to estimate review rounds.

Supplementary Figures 2–5 group additional evidence by assay. Main Figure 5 shows translocation and volume localisation; main Figure 4 leads with three blind treatment analyses and includes separate public and laboratory neurite demonstrations. The assisted mosaic is linked below as a development example. Reference evaluation follows pipeline selection; examples lacking exhaustive annotations do not receive an accuracy percentage.

- Bright-object analysis.
- Volume localisation.
- Paired-channel analysis.
- Retinal soma localisation.
- Personal nine-field mosaic.
- Independent field-by-field laboratory neurite analysis, with frozen pipeline and output identities, sampled raw/path review and remaining limitations.
- Public translocation assay.
- Public neurite analysis.
- BBBC039 frozen predictions on image-uninspected fields, with the field-level scores and exposure flags and source and image-opening evidence.
- BBBC013 endpoint selection and development-well inclusion.

- Trial wall times, usage and dated model/software identities, with the complete resource catalogue.

Two additional final-pipeline repeats appear in the trial-level inventory above. Bright-object detection analysis 2 selected 62 labels after matched image review; no computational-reference score was reported for that repeat. 3D nuclear localisation analysis 2 matched all 15 annotated centres within 30 voxels, with mean error 4.8580507660 voxels and eleven unmatched predictions. The original records linked above give their selected candidates and review scope. These repeats are separate from bright-object detection analysis 1 and 3D nuclear localisation analysis 1, plotted in main Figures 3 and 5.

The final nine-field pipeline is supplied unchanged. To reproduce it, relocate input/output destinations and the viewer endpoint, then validate, compile and execute through MCP with the recorded installation. The linked review gives the input manifest and review scope.

The following source files are also supplied byte-for-byte from the authors' selected frozen attempts. Their run identities and original hashes remain in the linked analysis records; no candidate was reselected using reference scores.

Analysis	Final pipeline
Bright-object detection 2	Source
3D nuclear localisation 2	Source
Paired nuclear/cell detection	Source
Retinal cell detection 2	Source
Public-image neurite tracing	Source
Protein translocation 2	Source
Nuclear detection 2	Source
Nuclear detection 3	Source
Cell-boundary analysis 3	Source

The two BBBC039 sources correspond to the plotted full-200 result (pooled F1 0.906) and the independent coverage repeat (0.898), respectively. Both require the original image metadata table; Nuclear detection analysis 3 also imports the supplied source binding. Relocate its original import directory and metadata location to these supplied files when reproducing the workflow. The binding matches the original source record for analysis 3 and the final input manifest for analysis 2. The final manifest for analysis 3 does not list the binding directly. The

BBBC007 source is the final analysis of all 16 paired fields; its relative `source_sets.csv` location refers to the supplied paired-field metadata.

The volume pipeline also requires the original centre-detection custom function. The translocation pipeline requires its original compartment measurements, assay statistics and dose-response summary functions. Register these sources through MCP under their recorded function names before validating the corresponding pipeline. Reproduction also requires the recorded inputs, metadata and software version. Preserve scientific settings while relocating destinations and use an isolated viewer endpoint. These are frozen analysis artifacts, not additions to the processing library. The retinal and volume sources in this table correspond to Data 8's named trials, not the different retinal trial in Supplementary Figure 4 or the volume trial in main Figure 5.

The personal mosaic and explicitly labelled development examples retain assisted provenance; they are not counted as fresh unguided trials. Recovery of principal neurite shafts is the relevant illustrated result, not exhaustive filopodial tracing or definitive neuron ownership at crossings.

Original attempts, unsuccessful candidates and delivery history remain in the linked records and the archived evidence inventory. They are not additional experimental replicates.

For NeuronCyto II image 1, the published manual-reference table contains eight traced-cell entries, without spatial identifiers or an exhaustive-coverage statement. The official testing archive contains the two source images but no manual spatial tracing files. Aggregate agreement therefore does not establish shaft recall, neuron ownership or calibrated length accuracy. The reference audit retains the source-table checksums, image correspondence and interpretation. The manual-reference construction is described by Ong et al. (2016).

The image-1 length comparison verifies that the H004 first prediction used the same source field. It reports 4,085.1925 pixels of rooted outgrowth; the published manual total is 3,832.601 in unspecified units. These values are descriptive, not an accuracy ratio: the length definitions and units have not been aligned, and the retained reference has no spatial traces or cell correspondences. The aggregate table keeps the first prediction separate from the overextended final attempted repair. Main Figure 4C–E illustrates the initial tracing rather than that later revision. The intended thick-shaft endpoint was clarified after the run; the original brief and all predictions remain unchanged. This clarification is not evidence that the original author received that more specific target. The primary-

source follow-through records the article and user-guide checks and the remaining unit requirement.

Additional paired-channel results

A further BBBC007 author analysed sixteen paired fields after self-directed repair, producing 1,428 nuclear and seeded-cell IDs. Saved masks reconcile reported areas and same-ID nucleus containment. The final-candidate analysis describes a possible remaining merge and crowded-boundary uncertainty.

An independent paired DNA/actin author recovered three crowded nuclear omissions and removed two candidates with no actin-supported growth beyond their nuclear seeds. Its final 56 nuclei and 54 cell regions have reconciled labels, areas and parent relationships. The completed analysis contains matched raw/label review and own-nucleus containment checks.

Analysis times, software versions and computational usage

The resource table records analysis times, software versions and token usage. The measurement definitions specify the timing boundaries and link original run identifiers. Analysis names below match the main text and Supplementary Data 8.

Time to first execution and task completion

Both times start when the agent was instructed to read the task brief. **First execution ended** marks the end of the first processing attempt, before subsequent image review; the attempt may have failed. **Task completed** marks the agent's completion report and includes corrections, image review, reporting and cleanup. It excludes subsequent reference scoring and archiving. For assisted development, the times cover only the recorded continuation. Minutes are rounded to two decimals.

Analysis	First execution ended (min)	Task completed (min)	First execution status
Bright-object detection 1	13.54	44.82	complete
3D nuclear localisation 1	20.20	43.33	failed

Analysis	First execution ended (min)	Task completed (min)	First execution status
Paired nuclear/cell detection	12.07	54.89	complete
Retinal cell detection 1	25.98	77.75	complete
Retinal cell detection 2	10.84	41.25	complete
Public-image neurite tracing	10.14	30.95	complete
Nuclear detection 2	21.28	69.14	failed
Nuclear detection 3	30.13	77.86	complete
Nuclear detection 4	23.66	174.62	failed
Cell-boundary analysis 3	18.83	90.76	complete
Protein translocation 2	18.10	213.10	complete
Laboratory-image neurite tracing	29.16	72.55	complete
Assisted nine-field continuation	8.66	54.62	complete

Some first executions failed after saving outputs: the 3D localisation run encountered a viewer error, and nuclear detection analysis 2 encountered a later metadata error. Other failed executions do not by themselves establish completed predictions or successful image review. Task duration includes development and review, so it is longer than a single execution of the final pipeline. Times for independent analyses and assisted development are reported separately.

Software used during the trials

The table groups recorded invocations and development phases by date, model, agent software and OpenHCS version. These dates identify trial records, not software releases. The 71 rows in the source table include continuations and repeated invocations and therefore do not represent 71 independent experiments. Source revisions, instruction-file checksums and mappings to the original sessions remain in the linked resource table.

Trial record date (UTC)	Model / provider	Agent software / OpenHCS	Recorded invocations or phases
2026-09-16	gpt-5.6-sol (paper-reported) / not recorded	not recorded / 0.8.5	3
2026-10-03	gpt-6.1-sol / openai	0.160.0 / 0.8.7	2
2026-10-04	gpt-6.1-sol / openai	0.160.0 / 0.8.7	21
2026-10-05	gpt-6.1-sol / openai	0.160.0 / 0.8.7	25
2026-10-06	gpt-6.1-sol / openai	0.160.0 / 0.8.7	20

For the three prospective September analyses, the paper records gpt-5.6-sol and OpenHCS 0.8.5, but original timing and usage logs were not located. Their task durations, agent software version and provider are therefore unavailable. Later records identify gpt-6.1-sol, provider label openai and agent software version 0.160.0. They do not identify an immutable backend model revision or establish which billing route was used.

Token usage and estimated costs

The primary resource measurement is recorded tokens, available for every retained invocation with counters in the resource CSV. The following selected rows correspond to the clock table above. M denotes million tokens and k thousand tokens; calculations use exact counters, not these rounded displays. Cached input is part of total input, not additional usage. Reasoning output is already included in output.

Trial	Input (M)	Cached input (M)	Output (k)	API scenario (USD)	Credit equivalent
Bright-object detection 1	16.95	16.49	45.0	3.03	75.7
3D nuclear localisation 1	20.64	20.16	47.1	3.45	86.4
Paired nuclear/cell detection	25.36	24.51	76.1	4.91	122.8
Retinal cell detection 1	18.93	18.35	67.6	3.67	91.8
Retinal cell detection 2	23.44	22.88	61.6	4.03	100.7

Trial	Input (M)	Cached input (M)	Output (k)	API scenario (USD)	Credit equivalent
Public-image neurite tracing	15.64	15.16	46.7	2.94	73.5
Nuclear detection 2	20.51	19.83	55.3	3.89	97.2
Nuclear detection 3	20.31	19.42	57.8	4.30	107.6
Nuclear detection 4	57.45	55.89	180.3	10.51	262.6
Cell-boundary analysis 3	34.48	33.37	105.3	6.60	165.1
Protein translocation 2	77.74	75.85	295.7	14.32	357.9
Laboratory-image neurite tracing	18.62	17.66	56.5	4.25	106.2
Assisted nine-field continuation	85.64	82.55	268.1	Not phase usage	Not phase usage

Rates and formulas are specified in the accounting reference, using the official API prices and Codex credit rates observed on 6 October 2026 for the exact recorded model. The API column is a Standard short-context read-token scenario, excluding unmeasured cache-write and other API fees; request-level context tiers have not been reconstructed. The credit column is a Standard-speed token equivalent, rather than a record of credits charged or subscription usage. These are scenario estimates, not actual historical charges.

Provider-billed USD remains `not_recorded` in the unchanged source CSV. Missing bills do not mean zero cost. Retained-context totals may include ancestors and are not incremental phase usage; the assisted row's token counters are reported but no phase-cost estimate is assigned. Older trials lacking counters cannot be priced from these records. Additional runs could supply prospective accounting, but are not required to report the existing usage.

The source table supplies the unrounded times, original usage counts and identifiers for every analysis above. The linked measurement definitions preserve the calculation details.

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